24/12/24

Unit-I

Introduction: Definition, Boundary, Psychological and Behavioural Influences on health and illness.
Neuroendocrine; transmitters & neuro-immune responses to stress, Negative affectivity, Behavioural patterns, Coping styles, Psychophysiological model of disease, theoretical model of health behaviour.

What is Behavioural Medicine?

Behaviour medicine is the inter-disciplinary field with the development and integration of behavioural, psycho-social, bio-medical science knowledge and techniques relevant to the understanding of health and illness.
The application of this knowledge and these techniques to prevention, diagnosis, treatment and rehabilitation.
B.M emphasizes remediation and healing of illness, whereas behavioural health focuses on prevention of disease.

The emergence of B.M

The emergence reflects the scientific recognition that health and disease are not biological phenomena but that how people act, think and feel a critically in determining who gets sick and how much sick they can

become and how well they can be treated.

Ancient Greek physicians and philosophers believed that mind and body are mutually inter-dependent and this interactionist approach was evident in medical practice as late as the reniassance period.

Greek physicians prescribed treatments such as relaxation inducing bath and massages and medieval remedies including listening to music, thinking comforting thoughts with objective resembling modern approaches to relaxation and stress management.

A number of 19th century physicians like Benjamin Rush, Sir William Oshler, Henry Holland have reintroduced the idea that both psychological and behavioural influences are also important for health and disease.

During 20th century, research from the field of psychosomatic medicine, behavioral and experimental psychology began to provide compeling evidence of mind & body or psychophysiological interaction.

1927 → Pavlov Classical conditioning "physiological responses can be influenced by non-biological cues"

Miller 1978, control of autonomic functions like heart rate, brain waves, body temperature and conditioning.

Cohen & Ader, 1988. Also demonstrated that immune function system could be conditioned to neutral stimuli.

1953, Wolff made important advances in the stress and adaptation in disease.

Alexander, 1953
Graham, 1972
Mason, 1975
} also achieved important insights with social, emotional and psychological and physical responses.

Behaviour Medicine term was first used in book by ~~Oscar~~ Lebirk, published in 1973.

⇒ 2 research unit. Centre of Behavioural Unit Ovide F. Pomerleau and John Paul Brady at University of Pennysdavania in 1973.

⇒ Another laboratory was founded by William Stewart Agras at Stanford University, 1974.

1977, Yale Conference on Behavioral Medicine Schwartz and Weiss proposed the bio-psychosocial model emphasizing the new roots of interdiscipline.

Scope of B.M

looks for improvement and longevity in the recognition of diseases of lifestyle.

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Models of Understanding Illness

- Social Cognition Models
  - Normative Beliefs
  - Parental Influence
  - Self Efficacy (I can stop smoking If I want to)
  - The concept of incentives (Behaviour is governed by the consequences)

The model gives importance to the cognition that differentiates between performing and not performing behaviours. So, in this model the focus is on cognition that will intervene between observable stimuli and real world situation behaviour.
— Fischk & Taylor 1991

It's based on the assumption that human behaviour is best understood as the function of people's perception of reality rather than objective categorisation of the stimulus environment.

Self-regulation research. Self-regulation processes are defined as mental and behavioural processes by which people
* enact their self conceptions.
* revise their behaviour
* alter the environment.
So as to bring outcomes according to

their perceptions and self-goals. Self-regulation helps the individual to eliminate dysfunctional pattern of thinking and behaviour & engage in adaptive patterns of thinking and behaviour.

It involves cognitive re-evaluation of beliefs, goal-setting, ongoing monitoring and evaluation of goal directed behaviour.

** 2 Phases of Self-Regulation Activities

1.) Motivational
2.) Volitional

Motivational phase: Analyzes cost & benefit and are considered to choose from goal and behaviour. This phase is assumed to conclude with a decision, or an intention concerning which goals and actions are pursued at a particular time.

Volitional phase: Planning an action, directed towards a set goal predominates.

The SCM majorly focuses on motivational but implementation & doing depends on volitional phase.

Protection Motivation Theory

→ Two sources of information: Environmental, Intrapersonal

1) Severity
2) Vulnerability
3) Response Effectiveness → Behavioural Intentions → Behaviour
4) Self-Efficacy
5) Fear

In PMT, the primary determinant is protection motivation or intention to perform a healthy behaviour.

2 appraisal process:

1. Threat appraisal
2. Coping appraisal

Threat appraisal: Is based on consideration of perception of susceptibility to illness, and severity of health threat.

Coping appraisal: Involves assessing the behavioural alternatives and reducing the threat.
It's based on 2 components:

1.) The individual's expectancy that carrying out a particular behaviour can remove the threat (action outcome efficacy).

2.) A belief in one's capability to successfully execute the recommended course of action (self-efficacy).

It will result either in adaptive responses or maladaptive responses.

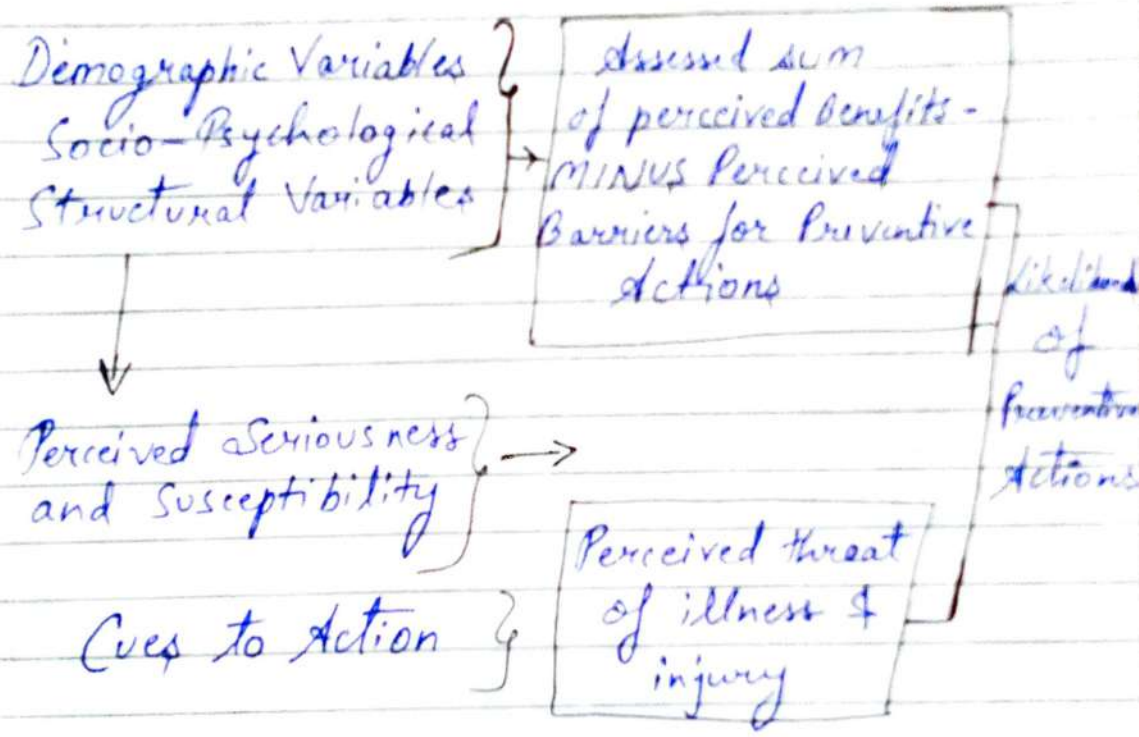
* Adaptive responses are those in which the individual engages in behaviour which are likely to reduce the risk, such as adopting a healthy behaviour.

* Maladaptive response are those that do not directly tackle the threat, there is denial of the threat.

Self-efficacy is one of the most important component of behaviour.

Health Belief Model

Perceptions & Modifying factors → Assessments → Likelihood of actions



The health model believes that health is defined by 2 cognition. It's one of the oldest & most widely used model.

1) Perception of Illness Threat
2) Evaluation of Behaviour to eradicate/counteract the threat

(1.) Threat perception are based on 2 belief.
a) The perceived susceptibility of the individual to the illness. How likely am I to get ill.

b.) Perceived severity of the consequences for the individual. How serious would the illness be.

(2.) Evaluation of Benefit:

1.) Consideration of potential benefit of

2.) Barrier / cost-to action.

Together these beliefs will determine how involved you are for health behaviour.

2 other cognition are also added in model:

1.) cues to Action
2.) Health Motivation

(1.) cues to action are assumed to include a diverse range of trigge to the individual taking action which may be internal like physical symptom or external i.e. mass-media campaign or advise from others to th individual.

An individual's perception of the presence of cues to expected to prompt adoption of the health behaviour, if the individual holds other key beliefs, favouring actions.

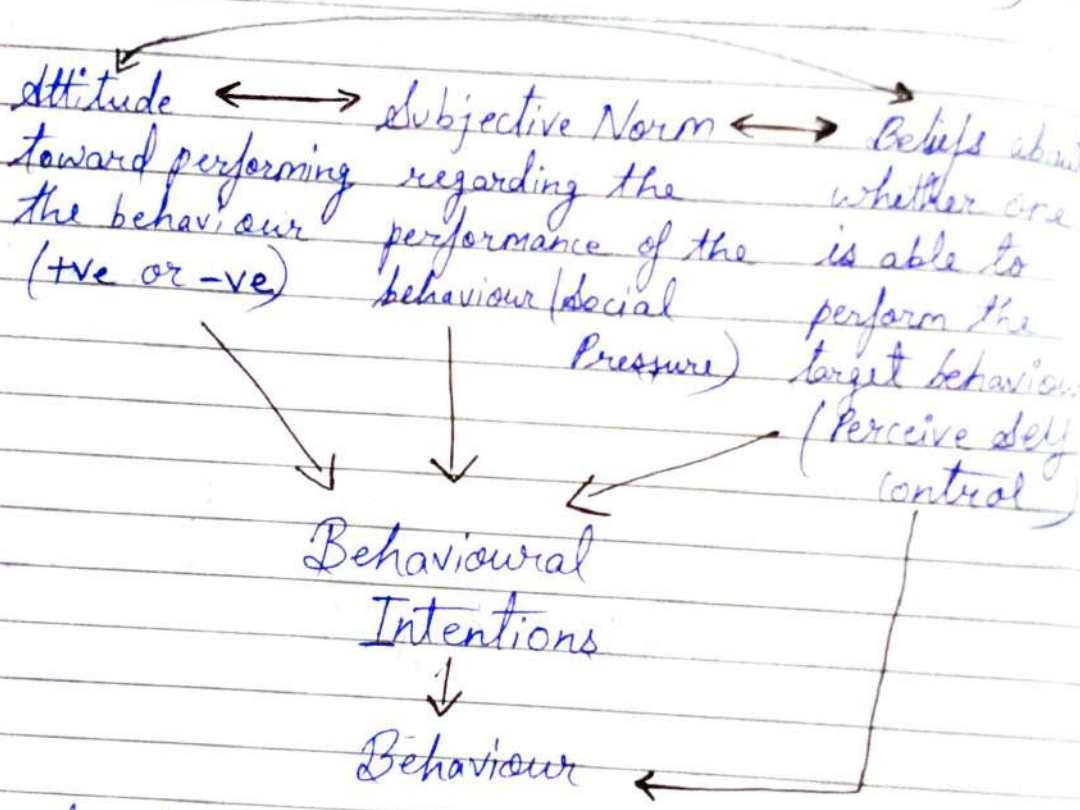
(2.) It refers to more stable references between individuals in the value they attach to their health and their propensity to be motivated to look for their health.

The main strength of HBM is the common sense operationalization. It uses including key believes related to decision about health behaviour.

This model is incomplete and can be improved.

Theory of Planned Behaviour

(Emphasis on Behaviour Intentions)



Specifies the factors that determines the individual decision to perform a particular behaviour, so it focuses on perceived behavioural control.

TPB proposes that the key determinants of behaviour are intention to engage in that behaviour and perceived control behavioural over that behaviour.

"The intentions in TPB represents a person's motivation or conscious plan or decision to exert effort to perform the behaviour.

(PBC)

Perceived Behaviour Control is a person's expectancy that the performance of the behaviour is within individual's control and confidence that he/she can perform the behaviour. Same as Bandura's Self-Efficacy.

Intention is determined by 3 factors:-

1.) Attitude toward performing behaviour.
2.) Subjective norm.
3.) Belief (PBC)

(1.) Our overall evaluation of the behaviour by the individual as +ve or -ve.
Attitudes are based on behavioural belief (outcome expectancy).
This expectancy framework is based on Fishbein's (1967) earlier summative model of attitudes.

(2.) Subjective norms are person's belief about whether significant others think that he/she should engage in that behaviour.
Salient others approval/disapproval.

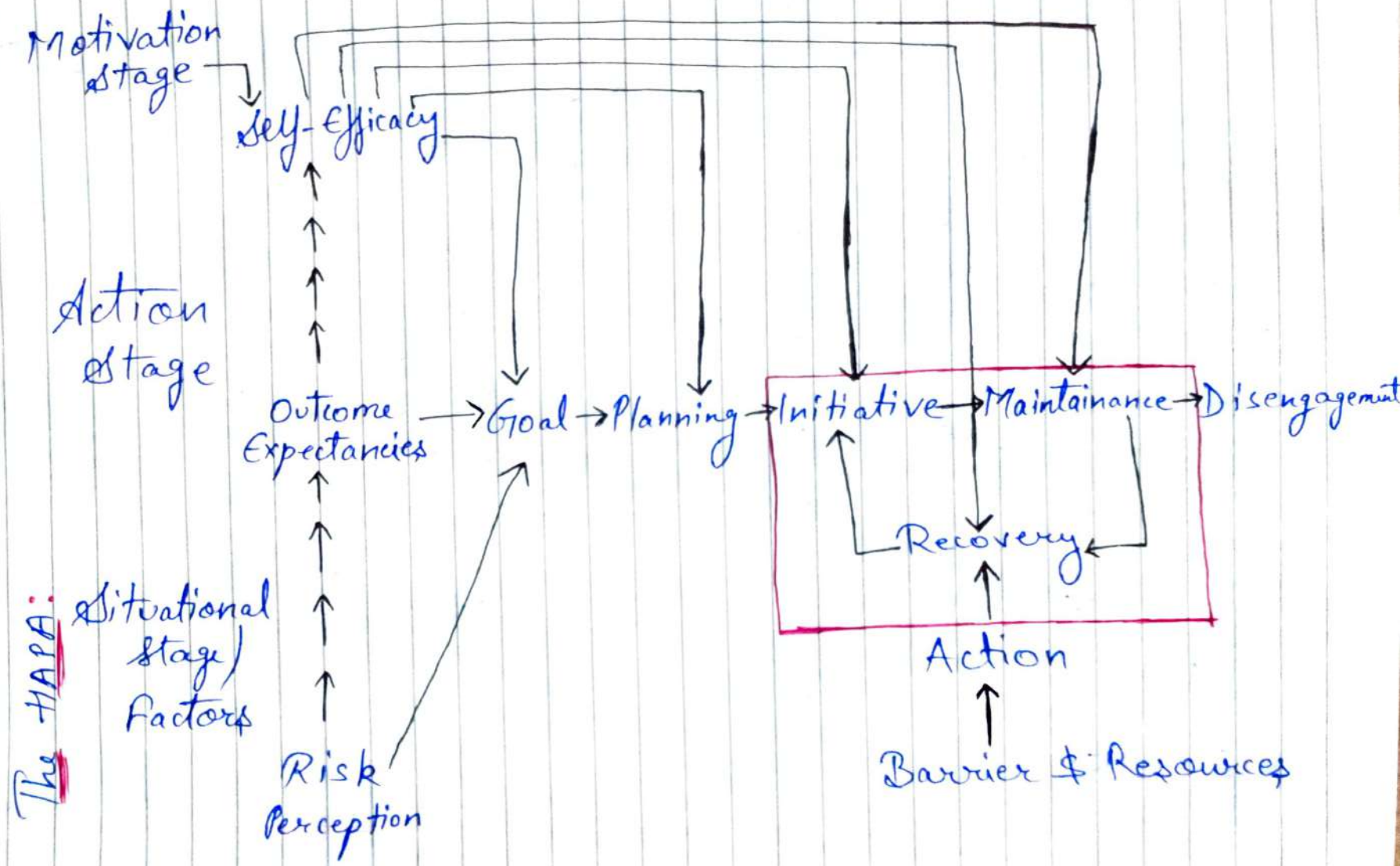
(3.) PBC is assumed to influence both intention & behaviour because we rarely intend to do things we know we cannot. & because believing that we can succeed enhances effort and persistance, so a successful performance

could happen. Therefore, PBC is based on controlled beliefs concerning whether one has access to the necessary resources and opportunities to perform the behaviour successfully, weighted by the perceived power of each factor to facilitate or inhibit the action, and these factors include both internal control factors such as 1) information 2) personal deficiencies 3) skills 4) abilities 5) emotions and external control factors such as 1) opportunities 2) dependence on others 3) barriers.

** Attitude is the strongest predictor among them. (2001)

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Health Action Process Approach (HAPA)



The HAPA:

Transtheoretical Model of Change

Not intending to make any change	Considering a change	Making a small change	Actively engage in a new behaviour
Pre-Contemplation	**Contemplation**	**Preparation**	**Action**
Change Process	• Dramatic Relief • Consciousness Raising • Help Relationship	Self Environment Re-evaluation	Social Liberation Self Stimulus control Counter condition Relationship
Self-Efficacy	Confidence Templation	Higher Confidence	Increasing Confidence Confidence Templation * Relapse can happen
Decision Balance	Cons > Pros	Cons = Pros	Cons ≥ Pros Cons < Pros

MAINTAINENCE
(self efficacy high) [sustaining change over time]

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Intention In all health behaviour model, there are some overlapping components -

Integration of Social Cognition Model

1.) Intention
2.) Self- Efficacy
3.) Outcome Expectancy

Therefore, there are certain finite set of variables to be considered in any behaviour analysis, & variables are accounted /identified.

1.) To have a strong intention.
2.) Have the necessary skill to perform the behaviour.
3.) Experience an absence of environmental constraint, that could prevent behaviour.

A strong intention has another 5 possibities:

1.) Individual perceives the advantages or benefits of performing the behaviour to outweigh the perceived disadvantages/ costs, i.e., outcome expectancies.

2.) Perceives the social, i.e. normative pressure to perform the behaviour to be greater than not to perform the behaviour.

3.) The individual beliefs that the behaviour is consistent with his/her self- image.

4.) Anticipates the emotional reaction to performing the behaviour to be more positive than negative.

5.) Have higher level of self-efficacy.

STRESS, HEALTH & COPING

According to Lazarus & Folkman, It is a transaction between person and the context.

Stress exists when a person confronts a circumstance that exceeds or taxes his/her ability to manage it. The circumstance itself is termed as "stressor".

It is Lazarus, the first person who emphasized that people's appraisal of the circumstance often matters more in the transaction than does the objective circumstance itself.

Appraisals are cognitive processes that not only incorporate information from the stressor but also information from inside the person.

Different type of appraisal for stress:

1.) Threat appraisal: Person views the stressor as an impending event that may have bad/harmful consequences.

2.) Harm appraisal: Something that has already happened bad.

3.) Loss appraisal: It's a specific kind of harm appraisal, the desired end becomes inaccessible.

4.) Challenge appraisal: Challenge here implies an optimal obstacle. It is demanding, difficult but appears to be over-mountable. The removal of obstacle will result in better state of well-being/living.

** → When your available resources are threatned, the utility of resources are critical then stress happens.

Physiological responses from stress

Stress response: It is a psychological, neural & hormonal changes in response to a stressor or an environmental stimulus that impacts an organisms' physiology. It can be considered as a series of counter-measures that they brain & body deploys inorder to cope with changes in Environment.

Therefore, stress responses promote adaptation to environment which is key to survival.

36.8°C
97-98°F

⇒ Homeostatis is maintaing right conditions for body & internal system.

⇒ Blood vessel constrict, goosebumps occur reducing heat dissipiation. Body's trying to maintain optimal level ⇒ In cold

⇒ Blood vessel dialate to increase heat dissipiation → Summer.

⇒ Body is maintained in it's optimal range because of interplay b/w neural, endocrine & exocrine system working together towards this single goal.

⇒ Maintaing homoestatis in the face of environmental changes is done because of process called allostatis.

Factors important for individual survivors:-

1) Temperature (2. mediater of allostatis (3. PH level (4. Oxygen tension (5. cortisol, epinephrine, cytokines)

All these factor ensures individual's survival in difficult or threatning situation to protect you for survival.

⇒ Elevated inflammatory cytokines can elevate cortisol → imbalance → sympathetic & parasympathetic

Allostatic state occur due to neuro-transmitter system & allevated level of circulating hormone that affect brain function & structure.

Brain tends to response through plasticity for having adaptive response, known adaptive plasticity.

↓

occur in dendrite & synaptic structure of neuron in hippocampus, amygdala and pre-frontal cortex, as one of the consequences of prolonged stress.

Adaptive is called coz they attempt to protect itself from cumulative damage from ~~&~~ stressor. But, if there is failure to do so, might lead to some irreparable damage & results in allostatic overload.

In states of chronic anxiety, depressed mood, neural damage, stroke, paralysis, neuro-degenerative diseases.

** ⇒ cost
Cause of allostatic overload can be significant.

An overly hypo-active, hypo-thalamic-pituatary-adrenal axis is equelly bad as a hyper-HPA axis.

*Catabolic

* CRH hormone :

HPA↑ → high gluco corticoids → ↑ cortico-tropin releasing hormone (CRH)

↓

catabolic activity → obesity

{hyper cortisol-emia}

** Hyper corticolemia → ↑ free fatty acid

insulin resistance → diabetes

visceral fat accumulation → sympathetic activation → hyper-tension

exacerbate ↑

⇒ Interaction of stress & HPA function and immune function as described as inverted U shape process.

⇒ Hypoactive HPA can lead to a hyper inflammatory immune state, specifically a hypoactive HPA is associated with an increase risk for chronic inflammation & also vulnerability to autoimmune-ity.

⇒ Chronic stress & exposure to high level of cortico-sterone dramatically reduce the delayed type hyper-sensitivity response.

The role of inflammation is clearly established in depression & depressive syndromes.

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Chapter 43

Physiological Responses of Stress

1) Confronting loss and threat produces emotional distress.

Emotional distress results in physiological changes like cardiovascular, neuro-endocrine changes, immune responses & in a state of emotional arousal like fear or anger, human body is ready for sustain action. The physiological changes will provide energy to body like muscles, digestion or self-repair.

& away from

⇒ Fear prepares body for escape, anger for attack.

Sadness/loss/rejection isn't about action preparing but giving up the efforts.

It is more imp. to devote resources to escape an immitent threat than to face an immediate infection.

BP → 120/80
systolic dystolic

Stressor responses are generally functionally.

⇒ Excessive cardiovascular system load causes abnormal high burden on "arteries". Over time, this creates small tear in the artery & there are depositing of protective plaques.
This is normal in small quantity. Too much plaque can result in atherosclerosis (clogging of artery).

⇒ Another ex, the process by which BP is regulated causes increase in baseline levels in response to situational demands. If those increases occur too frequently & are too sustained (बार बार रहते हैं), then the regulatory process starts to adjust the baseline upwards.
Now, when the stressor ends, the blood pressure returns not to its pre-existing level but to a higher level.
Over iterations (several adjustment) repeatition, the resting level becomes elevated. Thus, resulting into hyper-tension.
According to Sparrenberger chronic stress is cause of hyper-tension.

⇒ Another focal point in the stress related physiological response is responding through the hypothalamic – pituatry – adreno cortical axis (HPA).
This HPA plays imp. role in stress response.
HPA activation is reflected through increase in level of several hormone.
Example, catecholamine, epinephrine, norepinephrine & cortisol.

Cortisol is considered a particular imp. stress hormone.

Elevation of cortisol level can suppress different level/aspects of immune functioning. For eg, when strong demands are made on behaviour like fight/flight, then immune surveilliance takes lower priority & gets suppressed.

⇒ Immune system itself is a system target of analysis of stress related physiological responding.

Immune system is the body's main line of defense against disease agent ranging from bacteria —— cancer cells.
So, If immune fn gets impaired over a sustained period of time

the individual becoms more vulnerable to both opportunistic infectious agents & to agents of disease that have already been at work in body. The result may be either disease promotion or disease progression.

⇒ Nature of psychophysiological response pattern.

Stress → Acute/chronic

Short term responses are adaptive, Prolonged stress are dysfunctional.

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The Arrangement of Stress Response

The stress response is the complex process which is initiated by central and immune system. The CNS reacts against macroscopic threats & controls whole body response, whereas immune system reacts against microscopic threats targeting at epithelial or endothelial barrier.

"Infection & tissue damage"

Stress ⇒ The h-p-a axis is activated & vasopressin (Macro) prolactic & growth hormone are released. Corticotropin release hormone & vasopressin synergystically increase ACTH (adreno-cortico tropin releasing hormone) secretion.

ACTH induces conversion of cholestrol to cortisol, which in term cooperates with SNS (sympathetic nervous system) to prepare body for response by mobilization of energetic substraits.
Increase of intra-vascular volume & blood pressure enhancement.

Stress → (Micro) The initial signal is amplified by the flow of lymphokines, an activated cells & it stimulates central stress response which finally terminates system over-stimulation.
Immune response & tissue damage contribute to sytemic inflammatory respond syndrome development & all these inflammatory signals are transferred to the CNS by vagus nerve, & activate HPA axis.

After acute stress response when ACTH, prolactin, growth hormone & thyroid hormone are activated, the pulsatile & physiologic pattern of neurohormones concentration appears. But inadequate concentration of plasma neuro-hormones level can lead to acute failure and shock.

HP-Gonadal axis
HP- Thyroid axis

HP- Growth hormone Axis

The release of GH (somatotropin) is stimulated by neurons from the arcuate nucleus containing GH releasing factor & is inhibited by neurons from the periventricular nuclear containing GH-inhibiting factor.

Further, NT, neuropeptides & hormones affect GH release such as dopamine TRH (thyroid regulating hormone), neuropeptide-Y, sex hormone, gluco-corticoid hormone & thyroid hormone. Also, food related substances such as free fatty acids, glucose, amino acids they change GH regulation.

As per calorie intake changes, physical activity ↓ and stress.

GH stimulates the production of hepatic insulin-like-growth factor-1. In concert with IGF-1 both hormones regulate growth, somatic development, cell metabolism.

Vasopressinergic System

Also acts as a neurotransmitter in the brain, play a role in social behaviour such as territorial aggression, mate guarding and pair bonding.

To control circadian rhythm, thermo-regulation & release of ACTH. It helps to regulate blood pressure

Sympatho-Adrenal-Medullary System

Cannon (1914)

Adrenal medulla innervated by the splanchnic nerve, secrete the catecholamine epinephrine & norepinephrine directly into blood stream.

Epinephrine & norepinephrine increase heart rate, force of heart contraction & cardiac output, shift the blood supply to coronary arteries, skeletal muscle & brain, while blood supply of skin, Kidney & mucosa is diminished.

They reduce time for blood clotting through effects on blood platelets.

PSYCHO-IMMUNOLOGY

Involves the study of the interaction of the consciousness, CNS & the immune system. The immune system involves body's defense against infection & aberrant (deviant) self-division.

Research evidence has shown that CNS influences immune function & conversely can vice-versa.

Brain is normally the part of immuno-regulatory network.

HPA axis leads to down regulation of in response to stress of the immune system. Thoughts & emotions stressful may reach through axon to the hypothalamus projected from the limbic system.

Cortico tropin releasing factor (CRF) is produced in the hypothalamus under stress condition, acts on the anterior pituitary to form ACTH which in turn stimulates the production of cortico-steriods in the adrenal cortex. Under condition of acute stress cortico-steriods have immuno-suppressive effects on the lympho-reticular system & marked anti-allergic &

anti-inflammatory effects.

Definition:

⇒ It is the branch of medicine that studies the effect of psychological & social factors on the functioning of immune system.

⇒ The study of the connection b/w mind & the immune system.

CRF leads to release of catecholamines which themselves may produce changes in lymphocytes, monocytes & leukocytes functions. Opieds are also elevated through stress & they are generally reported to be immuno-suppressive.

Both GrH & prolactin which are immuno enhancing factor are initially activated at acute stress onset but condition of prolonged stress, this secretion is inhibited.

Thus, the combined effect of elevated Cortico-steriods, catecholamine, opieds along with inhibition of GrH & prolactin tends to dysregulate the immune system.

Recent evidence suggest that chronic psychological stress can lead to ↑ production of pro-inflammatory

cytokines, particularly interleukin (IL-6) which is triggered by infection & trauma.

Through inflammatory cytokines have been implicated in a range of disease in inflammation, cardiovascular disease, osteoporosis, Diabetes Type II, arthritis, certain lymphoproliferated diseases or cancer, multiple myeloma, chronic lymphocytic leukemia, alzeihmer disease.

IL-6 promotes the production of C-reactive protein, so it is an imp. factor MI (myocardial infarction). Depression & anxiety also enhances the production of pro-inflammatory cytokines.

⇒ ANS, HPA-axis

4 types of allostatic load can be result from prolonged stress.

1) Repeated alleviation of BP, over weeks or months accelerate atherosclerosis increases the risk of MI.

2) When an individual is unable to adapt to prolonged stress, he/she is exposed to stress hormone.

3) And if allostatic responses are not dealt correctly, it can lead to ↑ hypertension, decrease bone density.

4) Inadequate responses such as cortisol secretion which in turn lead to abnormal increase in pro-inflammatory cytokines which are in turn regulated by cortisol.

The specific immune system can be divided into 2 functional categories

1) circulatory measures
2) illicted measures

(1) Circulatory Measures consist of natural Killer cells

I → Stress: Neurobiology Consequences & Management.

II - Glutamate: The master NT & it's implication (UK.) in chronic stress & mood disorder.

2/01/2025

Psycho-neuro-Immunology

PNI examines how psychological, neural and immunologic processes interact & influence e/o, shape human health and behaviour

PNI involves the discovery that the components of immune system involved in inflammation are influenced not only by virus $ infection present inside the body but also by the cues, signals $ events occuring in external environment.
Research have shown that social-environmental processes, including psychological stress, can substantially upregulate inflammatory activity.

Immune System

The goal of human immune system is to keep the body biologically safe $ protected from foreign pathways.

It has 2 inter-connected branches referred to a.) innate immunity and b.) adaptive immunity. These branches work together to provide humans with short and long-term protection against pathogens that could enter the body.

Innate Immunity

The innate immunity represents the body's highly conserved, rapid, first-line defense against tissue damage and microbial infection.

The innate immune system activation begins when receptors located on immune cells recognize highly conserved features of microbes or pathogen-associated molecular patterns (PAMPs).

The recognition strategy is called pattern-recognition through pattern-recognition receptors.

One of the most well-characterized pattern-recognition receptors are toll-like receptors (TLRs).

TLRs are present on macrophages, neutrophils, dendritic cells.

⇒ When innate immune system defenses are insufficient for addressing a biological threat, adaptive immunity is called into action.

AI involves proliferation of microbial-specific white blood cells (lymphocytes).

AI is initiated by antigen-presenting cells (APCs), such as macrophages or dendritic cells, which help immune system differentiate b/w the host's own cell & those of invading bacteria (foreign).

COPING

17/01/25

Coping is generally defined as an effort to deal in some manner with a threatning or harmful situation, either to diminish the threat or remove the affects in which it can have adverse effect on the person.

Lazarus & Folkman 1984, talked about problem-focused coping or emotional-focused coping.

⇒ Problem-focused coping is aimed at the stressor itself. It may involve taking steps to remove or evade the stressors' arrival, Or to reduce it's physical impact.

⇒ Emotion-focused coping stems from the fact that stress emerges typically incorporate distressed emotions. It is aimed at reducing distress in one fashion or the other.

Problem focused coping strategies are well suited to stressors that seemed to have controllable aspects, whereas emotion focused coping strategies better suited to stressors that the person seems are uncontrollable.

- Park et al; 2004

The goal of the 2 copings are different.

When people engage in problem solving strategies they indirectly reduce their emotional distress whereas when people engage in emotion focused strategies may therefore be able to face a problem more calmly and generate a more effective problem focused strategies.

imp
Another very distinction among coping response is between approach and avoidance and engagement - disengagement.
- Skinner et al, 2003

Approach coping strategies are efforts to deal with the stressor or related emotions.
→ short-term ke liye thik.

Avoidance strategies are attempt to escape from having to deal with stressors.

Disengagement coping is also regarded as emotion focused style coz it involves an attempt to escape from feeling of distress.
Sometimes this coping is almost literally an effort so as to act as the stressor does not exist, so it does not have to be reacted to.

Wishful thinking and fantasy distances the person from the stressor, atleast temporarily and denial then creates a boundaries b/w from personal experiences & reality.

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Suicidal Assessment

I → Ideation
S → Substance abuse
P → Purposelessness
A → Anxiety
T → Trapped
H → Hopelessness
W → Withdrawal
A → Anger
R → Recklessness
M → Mood change

Difference between Myth & Facts

Myths	Facts
(1.) Once suicidal, always suicidal.	(1.) Short-term and situation specific.
(2.) Using word "suicide" induces suicidal ideation.	(2.) Talking openly gives the person an option to discuss.

3. Occurs only for person with mental illness.	3. Mostly situational specific.
4. Happens without warning.	4. Warning signs are present, either verbal or behaviour.
5. Someone who is suicidal is determined to die.	5. They are ambivalent about dying or living, hence emotional support will help them.
6. People who talk about suicide do not mean to do it.	6. Expression of anxiety & depression are present.

Suicide Risk Stratification

Low: Occasional wish to die but no actual plan to die.

Medium: Active plan but no attempt.

High: Recent suicide attempt and active plan.